

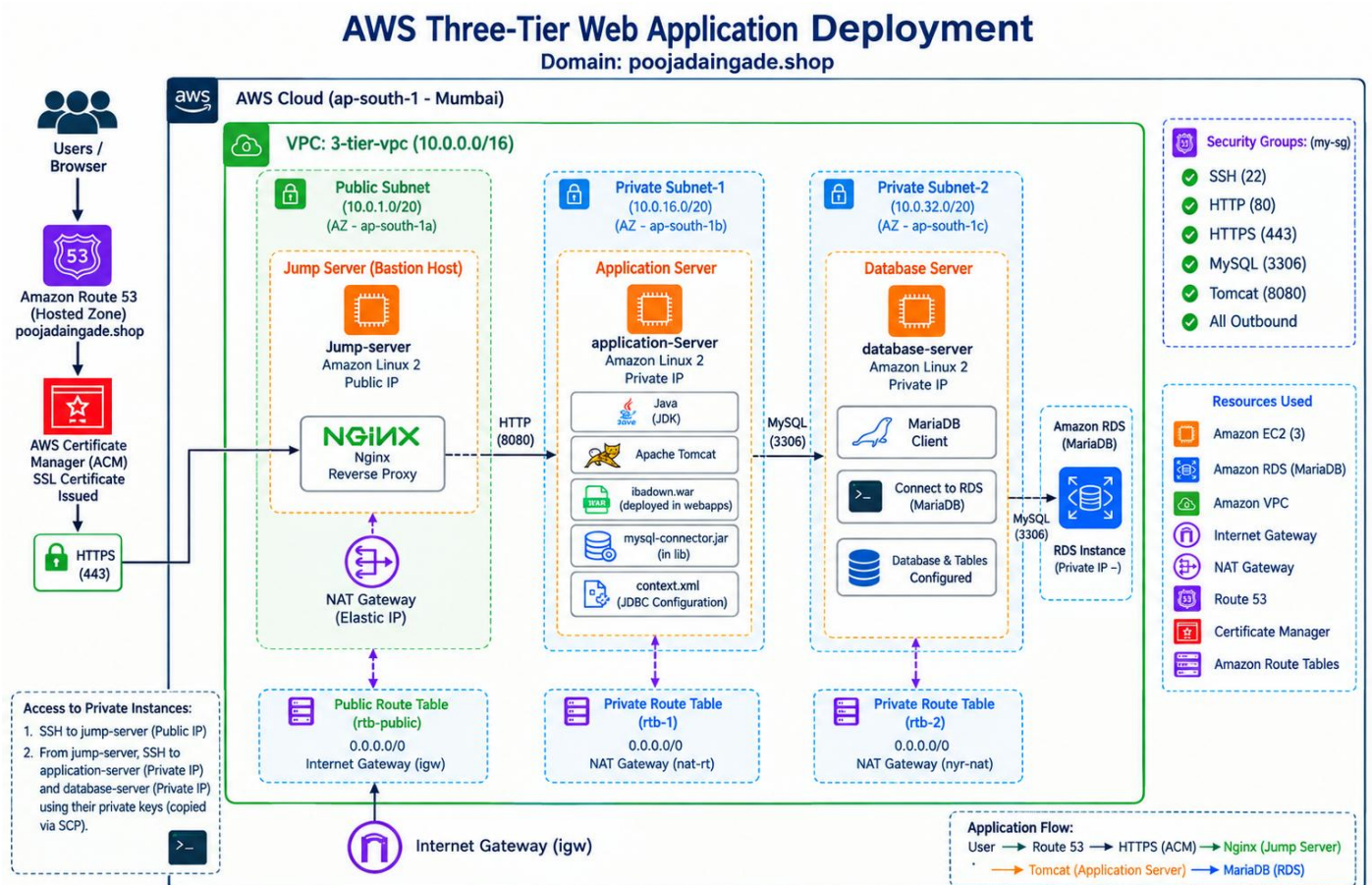
AWS Three-Tier Web Application Deployment

Prepared By: Pooja Daingade

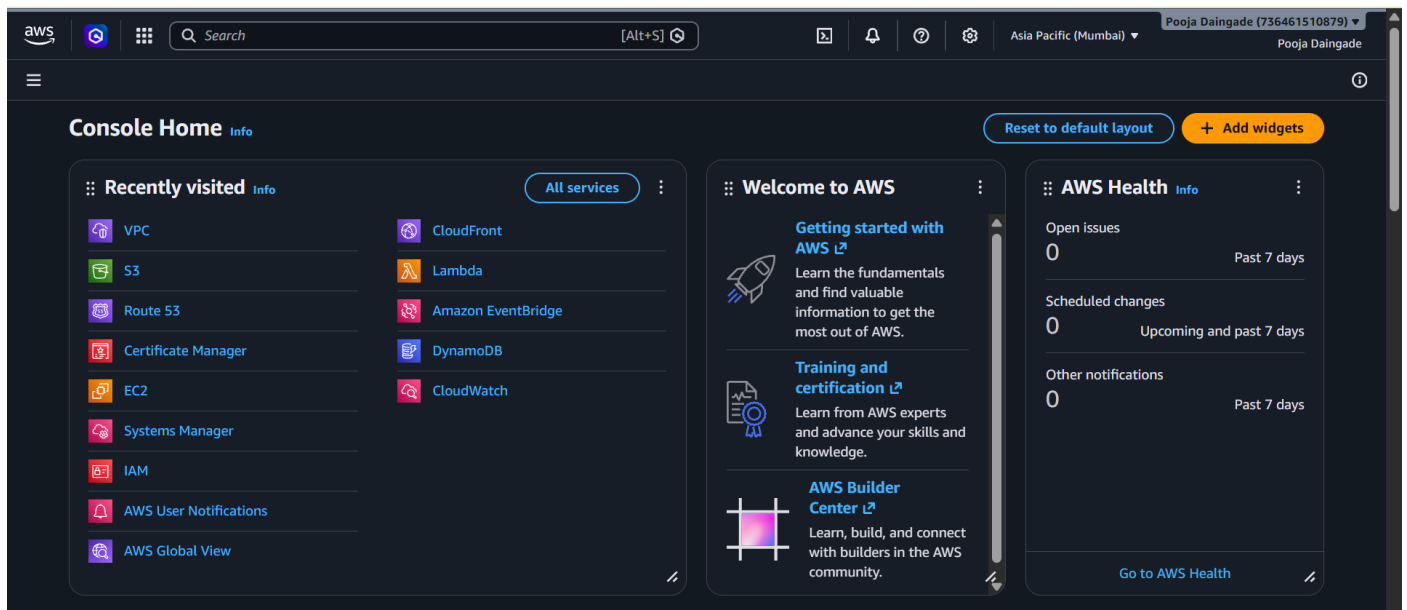
Project Overview

Deployed a Three-Tier Web Application on AWS using Amazon VPC, EC2, Nginx, Apache Tomcat, RDS MariaDB, Route 53, and ACM. The architecture separates the Web, Application, and Database layers across public and private subnets. Nginx was configured as a reverse proxy to forward requests to the Apache Tomcat application server, which connects to RDS MariaDB for database operations. The project also included custom domain configuration and HTTPS/SSL, with successful end-to-end application and database connectivity verification.

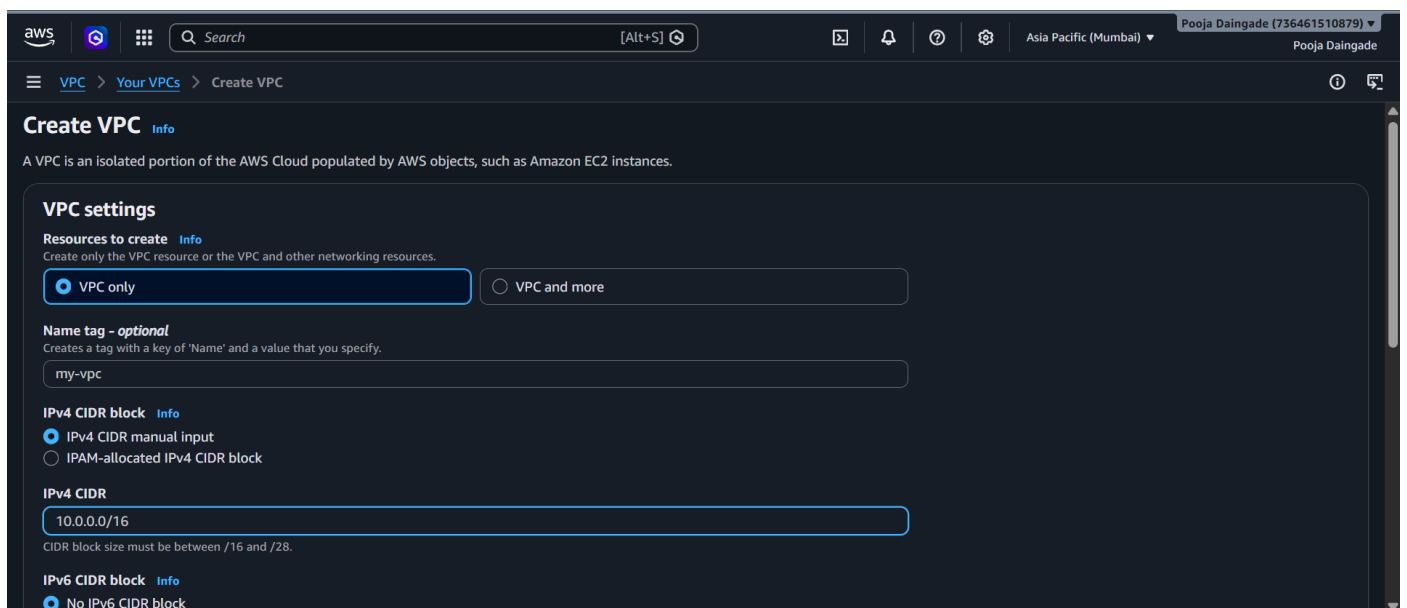
Architecture Diagram



1. Logged in to the AWS Management Console to access AWS services required for the project. From the Console Home, services like EC2, VPC, Route 53, ACM, RDS were used to deploy and manage the Three-Tier Application infrastructure.



2. Created a custom **Virtual Private Cloud (VPC)** named **my-vpc** with the IPv4 CIDR block **10.0.0.0/16** to provide an isolated and secure network environment for deploying the Three-Tier Application infrastructure.



3. Created three subnets within the **my-vpc** to separate different application layers. A **public-subnet** was created for internet-facing resources, while **private-subnet-1** and **private-subnet-2** were created for securely hosting the application and database servers.

aws    Search [Alt+S]     Asia Pacific (Mumbai) Pooja Daingade (736461510879) Pooja Daingade

VPC > Subnets > Create subnet

Create subnet [Info](#)

VPC

VPC ID
Create subnets in this VPC.

vpc-06eb8795e39f55c88 (my-vpc)

Associated VPC CIDRs

IPv4 CIDRs
10.0.0.0/16

VPC > Subnets > Create subnet

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.

public-subnet

The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.

Asia Pacific (Mumbai) / ap-south-1a

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16

IPv4 subnet CIDR block

10.0.0.0/20 4,096 IPs

Subnet 2 of 2

Subnet name
Create a tag with a key of 'Name' and a value that you specify.

private-subnet-1

The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.

Asia Pacific (Mumbai) / ap-south-1b

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16

IPv4 subnet CIDR block

10.0.16.0/20 4,096 IPs

Subnet 3 of 3

Subnet name
Create a tag with a key of 'Name' and a value that you specify.

private-subnet-2

The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.

Asia Pacific (Mumbai) / ap-south-1c

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16

IPv4 subnet CIDR block

10.0.32.0/20 4,096 IPs

4. Enabled **Auto-assign Public IPv4 address** for the public-subnet to automatically assign a public IPv4 address to EC2 instances launched in this subnet.

The screenshot shows the AWS Management Console interface. The top navigation bar includes the AWS logo, a search bar, and the region 'Asia Pacific (Mumbai)'. The left sidebar shows the 'VPC' dashboard with options to filter by VPC, view VPCs, subnets, and route tables. The main content area displays 'Subnets (1/3)' for VPC 'vpc-06eb8795e39f55c88'. A table lists subnets: 'public-subnet' (ID: subnet-04bb543481b00c19c), 'private-subnet-1' (ID: subnet-069444d44295b5801), and 'private-subnet-2' (ID: subnet-0bd9478524583bd2d). An 'Actions' menu is open for the 'public-subnet', showing options like 'View details', 'Create flow log', 'Edit subnet settings', 'Edit IPv6 CIDRs', 'Edit network ACL association', 'Edit route table association', 'Edit CIDR reservations', 'Share subnet', 'Manage tags', and 'Delete subnet'. Below this, the 'Edit subnet settings' page is shown for 'subnet-04bb543481b00c19c'. It displays the subnet ID and name, and the 'Auto-assign IP settings' section where the checkbox 'Enable auto-assign public IPv4 address' is checked.

5. Created an Internet Gateway (IGW) and attached it to the **my-vpc** to enable internet connectivity for resources deployed in the public subnet.

The screenshot shows the AWS Management Console interface. The top navigation bar includes the AWS logo, a search bar, and the region 'Asia Pacific (Mumbai)'. The left sidebar shows the 'VPC' dashboard with options to filter by VPC, view VPCs, subnets, and route tables. The main content area displays 'Create internet gateway' for VPC 'vpc-06eb8795e39f55c88'. The 'Internet gateway settings' section shows the 'Name tag' as 'my-igw'. The 'Tags - optional' section shows a tag with key 'Name' and value 'my-igw'. Below this, the 'Internet gateways (1/2)' page is shown. A table lists internet gateways: 'my-igw' (ID: igw-02ead48d41e72d8b8). An 'Actions' menu is open for 'my-igw', showing options like 'View details', 'Attach to VPC', 'Detach from VPC', 'Manage tags', and 'Delete internet gateway'.

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VPC > Internet gateways > Attach to VPC (igw-02ead48d41e72d8b8)

Attach to VPC (igw-02ead48d41e72d8b8) Info

VPC
Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.

Available VPCs
Attach the internet gateway to this VPC.

Q select a VPC

vpc-06eb8795e39f55c88 - my-vpc

Cancel Attach internet gateway

- Created a Route Table named **public-rt** and added a route (0.0.0.0/0) pointing to the Internet Gateway (IGW). The route table was then associated with the **public-subnet** to provide internet access for resources deployed in the public subnet.

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VPC > Route tables > Create route table

Create route table Info

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

public-rt

VPC
The VPC to use for this route table.

vpc-06eb8795e39f55c88 (my-vpc)

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key **Value - optional**

Q Name X Q public-rt X Remove

Add new tag

You can add 49 more tags.

Cancel Create route table

Edit routes: rtb-0cf18a1ea0f4b942e / public-rt

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
Q 0.0.0.0/0 X	Internet Gateway	-	No	CreateRoute
	igw-02ead48d41e72d8b8			
	Use: "igw-02ead48d41e72d8b8"			
	igw-02ead48d41e72d8b8 (my-igw)			

Add route Remove

Cancel Preview Save changes

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/3)

Q Filter subnet associations

	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☒	public-subnet	subnet-04bb543481b00c19c	10.0.0.0/20	-	Main (rtb-0209400faa8259a72)
☐	private-subnet-1	subnet-069444d44295b5801	10.0.16.0/20	-	Main (rtb-0209400faa8259a72)
☐	private-subnet-2	subnet-0bd9478524583bd2d	10.0.32.0/20	-	Main (rtb-0209400faa8259a72)

Selected subnets

subnet-04bb543481b00c19c / public-subnet X

Cancel Save associations

7. Created a NAT Gateway named **my-nat** in the **Public Subnet** and associated it with an **Elastic IP Address**.

The screenshot shows the 'Create NAT gateway' page in the AWS Management Console. The page is titled 'Create NAT gateway' with an 'Info' icon. Below the title is a description: 'A highly available, managed Network Address Translation (NAT) service that instances in private subnets can use to connect to services in other VPCs, on-premises networks, or the internet.'

NAT gateway settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.
my-nat
The name can be up to 256 characters long.

Availability mode **Info**
Choose whether to deploy across all zones in the region or restrict to a single availability zone.

☐ **Regional - new**
Scales automatically across all regional AZs, simplifying management for multi AZ deployments.

☒ **Zonal**
Provides granular control within a specific availability zone, adhering to subnet level settings.

Subnet
Select a subnet in which to create the NAT gateway.
subnet-04bb543481b00c19c (public-subnet)

Connectivity type
Select a connectivity type for the NAT gateway.
☒ **Public**
☐ **Private**

Elastic IP allocation ID **Info**
Assign an Elastic IP address to the NAT gateway.
eipalloc-00c4e3a91af9caed2 **Allocate Elastic IP**

8. Created a NAT Route Table named **nat-rt** and added a default route (0.0.0.0/0) pointing to the **my-nat** NAT Gateway. The route table was then associated with **private-subnet-1** and **private-subnet-2** to provide secure outbound internet access for instances in the private subnets.

The screenshot shows the 'Create route table' page in the AWS Management Console. The page is titled 'Create route table' with an 'Info' icon. Below the title is a description: 'A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.'

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.
nat-rt

VPC
The VPC to use for this route table.
vpc-06eb8795e39f55c88 (my-vpc)

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key
Q Name X

Value - optional
Q nat-rt X Remove

Add new tag
You can add 49 more tags.

Create route table

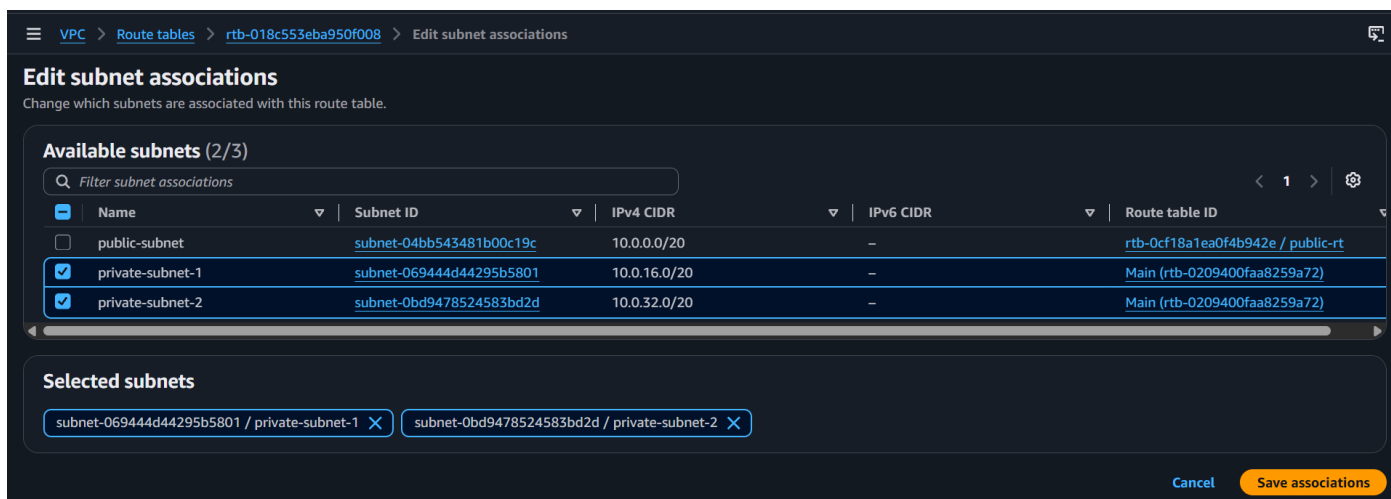
The screenshot shows the 'Edit routes' page for route table 'rtb-018c553eba950f008' in the AWS Management Console. The page is titled 'Edit routes: rtb-018c553eba950f008 / nat-rt'.

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	NAT Gateway	-	No	CreateRoute

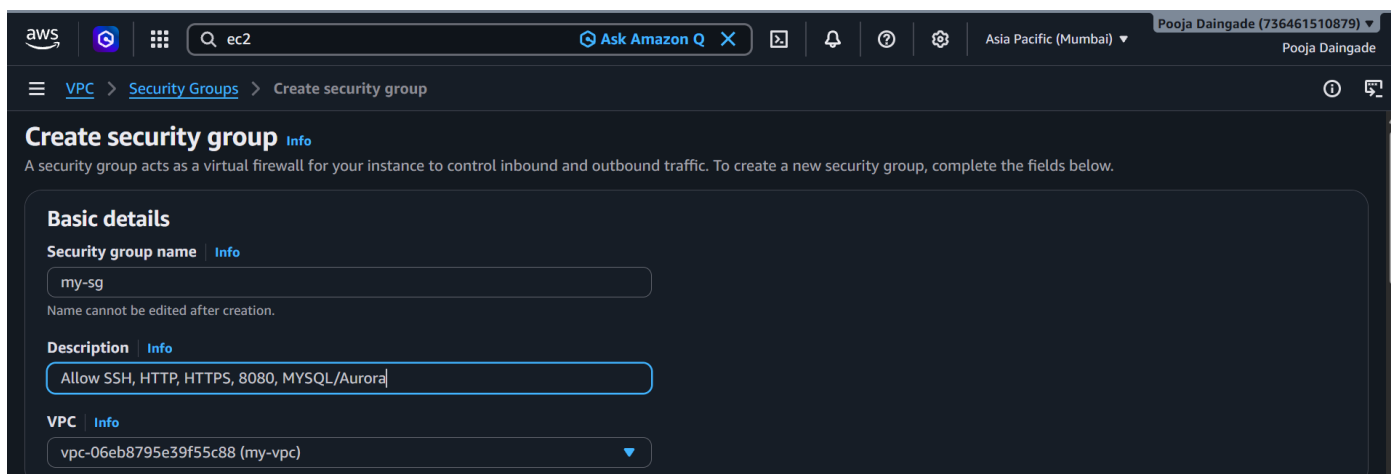
Add route

nat-01142cd8962682ff0 (my-nat)

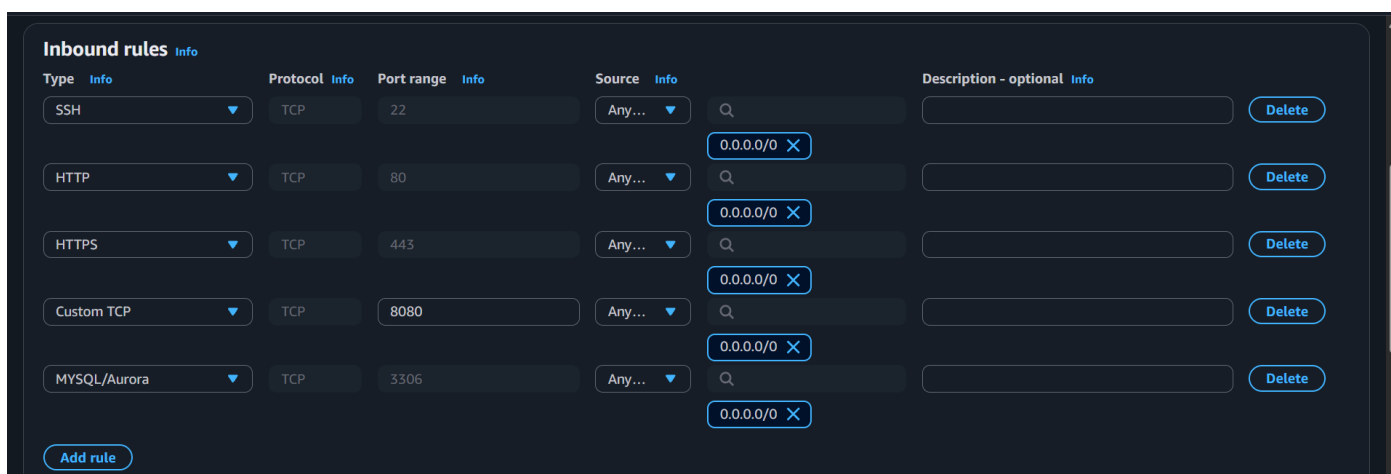
Save changes



- Created a Security Group named **my-sg** within the **my-vpc** to control inbound and outbound traffic for the project resources.



- Configured the inbound rules for my-sg to allow the required traffic for the Three-Tier Application. The following ports were allowed: **SSH (22)**, **HTTP (80)**, **HTTPS (443)**, **MySQL/Aurora (3306)**, and **Custom TCP (8080)**.



11. Launch EC2 Instances - 1

Launched an Amazon EC2 instance named **jump-server** using the **Amazon Linux** AMI and **t3.micro** instance type. The instance was deployed in the **public-subnet** of **my-vpc**, associated with the **my-sg** security group, and configured with the **3-tier-key** key pair to provide secure SSH access to the private infrastructure.

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name: [Add additional tags](#)

Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Recents **Quick Start**

Amazon Linux macOS Ubuntu Windows Red Hat SUSE Linux Debian

[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.18 AMI
ami-08e3b3155fc937a94 (64-bit (x86) - uefi-preferred) / ami-0cf36b0c962e50fd7 (64-bit (arm) - uefi)

Summary

Number of instances [Info](#)

1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.12....[read more](#)
ami-08e3b3155fc937a94

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

Instance type [Info](#) [Get advice](#)

Instance type

t3.micro [Free tier eligible](#)

Family: t3 2 vCPU 1 GiB Memory Current generation: true

On-Demand Linux base pricing: 0.0112 USD per Hour On-Demand SUSE base pricing: 0.0112 USD per Hour

On-Demand Windows base pricing: 0.0204 USD per Hour On-Demand Ubuntu Pro base pricing: 0.0147 USD per Hour

On-Demand RHEL base pricing: 0.04 USD per Hour

[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

3-tier-key [Create new key pair](#)

Summary

Number of instances [Info](#)

1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.12....[read more](#)
ami-08e3b3155fc937a94

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Network settings [Info](#)

VPC - required [Info](#)

vpc-06eb8795e39f55c88 (my-vpc)
10.0.0.0/16

Subnet [Info](#)

subnet-04bb543481b00c19c public-subnet [Create new subnet](#)

VPC: vpc-06eb8795e39f55c88 Owner: 736461510879
Availability Zone: ap-south-1a (aps1-az1) Zone type: Availability Zone
IP addresses available: 4091 CIDR: 10.0.0.0/20

Auto-assign public IP [Info](#)

Enable

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group ☒ Select existing security group

Common security groups [Info](#)

Select security groups

my-sg sg-06e6d9cfc9a9a043d [X](#)

VPC: vpc-06eb8795e39f55c88

[Compare security group rules](#)

Security groups that you add or remove here will be added to or removed from all your network interfaces.

Summary

Number of instances [Info](#)

1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.12....[read more](#)
ami-08e3b3155fc937a94

Virtual server type (instance type)
t3.micro

Firewall (security group)
my-sg

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

12. Launch EC2 Instances - 2

Launched an Amazon EC2 instance named **application-server** using the **Amazon Linux** AMI and **t3.micro** instance type. The instance was deployed in **private-subnet-1** of **my-vpc**, associated with the **my-sg** security group, and configured with the **3-tier-key** key pair to securely host the application.

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name
application-server [Add additional tags](#)

Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Recents **Quick Start**

Amazon Linux macOS Ubuntu Windows Red Hat SUSE Linux Debian

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.18 AMI
ami-08e3b3155fc937a94 (64-bit x86_64, uefi, preferred) / ami-0c36b0c962e50fd7 (64-bit Arm, uefi)

Summary

Number of instances [Info](#)
1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.12....[read more](#)
ami-08e3b3155fc937a94

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

Instance type [Info](#) [Get advice](#)

Instance type
t3.micro [Free tier eligible](#)

Family: t3 2 vCPU 1 GiB Memory Current generation: true
On-Demand Linux base pricing: 0.0112 USD per Hour On-Demand SUSE base pricing: 0.0112 USD per Hour
On-Demand Windows base pricing: 0.0204 USD per Hour On-Demand Ubuntu Pro base pricing: 0.0147 USD per Hour
On-Demand RHEL base pricing: 0.04 USD per Hour

☐ All generations [Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*
3-tier-key [Create new key pair](#)

Summary

Number of instances [Info](#)
1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.12....[read more](#)
ami-08e3b3155fc937a94

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

Network settings [Info](#)

VPC - required [Info](#)
vpc-06eb8795e39f55c88 (my-vpc) 10.0.0.0/16

Subnet [Info](#)
subnet-069444d44295b5801 private-subnet-1 [Create new subnet](#)

VPC: vpc-06eb8795e39f55c88 Owner: 736461510879
Availability Zone: ap-south-1b (aps1-az3) Zone type: Availability Zone
IP addresses available: 4091 CIDR: 10.0.16.0/20

Auto-assign public IP [Info](#)
Disable

Firewall (security groups) [Info](#)
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group ☒ Select existing security group

Common security groups [Info](#)
Select security groups [Compare security group rules](#)

my-sg sg-06e6d9cfc9a9a043d [X](#)
VPC: vpc-06eb8795e39f55c88

Security groups that you add or remove here will be added to or removed from all your network interfaces.

Summary

Number of instances [Info](#)
1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.12....[read more](#)
ami-08e3b3155fc937a94

Virtual server type (instance type)
t3.micro

Firewall (security group)
my-sg

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

13. Launch EC2 Instances - 3

Launched an Amazon EC2 instance named **database-server** using the **Amazon Linux AMI** and **t3.micro** instance type. The instance was deployed in **private-subnet-2** of **my-vpc**, associated with the **my-sg** security group, and configured with the **3-tier-key** key pair for secure database administration.

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name
database-server [Add additional tags](#)

Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Recents **Quick Start**

Amazon Linux macOS Ubuntu Windows Red Hat SUSE Linux Debian

[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.18 AMI
ami-08e3b3155fc937a94 (64-bit (x86), uefi-preferred) / ami-0c336b0c962e50fd7 (64-bit (Arm), uefi)

Summary

Number of instances [Info](#)
1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.12....[read more](#)
ami-08e3b3155fc937a94

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

Instance type [Info](#) [Get advice](#)

Instance type

t3.micro [Free tier eligible](#)

Family: t3 2 vCPU 1 GiB Memory Current generation: true

On-Demand Linux base pricing: 0.0112 USD per Hour On-Demand SUSE base pricing: 0.0112 USD per Hour

On-Demand Windows base pricing: 0.0204 USD per Hour On-Demand Ubuntu Pro base pricing: 0.0147 USD per Hour

On-Demand RHEL base pricing: 0.04 USD per Hour

[All generations](#) [Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

3-tier-key [Create new key pair](#)

Summary

Number of instances [Info](#)
1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.12....[read more](#)
ami-08e3b3155fc937a94

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Network settings [Info](#)

VPC - required [Info](#)

vpc-06eb8795e39f55c88 (my-vpc) [Refresh](#)

10.0.0.0/16

Subnet [Info](#)

subnet-0bd9478524583bd2d private-subnet-2 [Create new subnet](#)

VPC: vpc-06eb8795e39f55c88 Owner: 736461510879

Availability Zone: ap-south-1c (aps1-az2) Zone type: Availability Zone

IP addresses available: 4090 CIDR: 10.0.32.0/20

Auto-assign public IP [Info](#)

Disable

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group ☒ Select existing security group

Common security groups [Info](#)

Select security groups [Compare security group rules](#)

my-sg sg-06e6d9cfc9a9a043d [X](#)

VPC: vpc-06eb8795e39f55c88

Security groups that you add or remove here will be added to or removed from all your network interfaces.

Summary

Number of instances [Info](#)
1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.12....[read more](#)
ami-08e3b3155fc937a94

Virtual server type (instance type)
t3.micro

Firewall (security group)
my-sg

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

14. Create Amazon RDS Database

Created an Amazon RDS **MariaDB** instance using the **Full Configuration** method. The database instance was configured with the identifier **database-1**, username **admin**, and deployed inside **my-vpc**. A new DB Subnet Group was created to host the database in the private subnets for secure access.

Create database [Info](#)

Free plan has access to limited features and resources. The free plan limits the features and resources that are available for RDS and Aurora databases. Upgrade your account plan to remove all limitations. [Learn more](#)

Engine options [Info](#)

Engine type

- ☐ Aurora (MySQL Compatible)
- ☐ Aurora (PostgreSQL Compatible)
- ☐ MySQL
- ☐ PostgreSQL
- ☒ MariaDB
- ☐ Oracle
- ☐ Microsoft SQL Server
- ☐ IBM Db2

Choose a database creation method

- ☒ Full configuration
You set all of the configuration options, including ones for availability, security, backups, and maintenance.
- ☐ Easy create
Use recommended best-practice configurations. Some configuration options can be changed after the database is created.

Engine version

MariaDB 11.8.8

DB instance identifier [Info](#)

Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.

database-1

The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 63 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

Credentials Settings

Master username [Info](#)

Type a login ID for the master user of your DB instance.

admin

1 to 16 alphanumeric characters. The first character must be a letter.

Credentials management

You can use AWS Secrets Manager or manage your master user credentials.

- ☐ Managed in AWS Secrets Manager - *most secure*
RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.
- ☒ Self managed
Create your own password or have RDS create a password that you manage.

☐ Auto generate password
Amazon RDS can generate a password for you, or you can specify your own password.

Master password [Info](#)

Password strength [Average](#)

Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / * @

Confirm master password [Info](#)

Connectivity [Info](#)

Compute resource

Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

- ☒ Don't connect to an EC2 compute resource
Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.
- ☐ Connect to an EC2 compute resource
Set up a connection to an EC2 compute resource for this database.

Network type [Info](#)

To use dual-stack mode, make sure that you associate an IPv6 CIDR block with a subnet in the VPC you specify.

- ☒ IPv4
Your resources can communicate only over the IPv4 addressing protocol.
- ☐ Dual-stack mode
Your resources can communicate over IPv4, IPv6, or both.

Virtual private cloud (VPC) [Info](#)

Choose the VPC. The VPC defines the virtual networking environment for this DB instance.

my-vpc (vpc-06eb8795e39f55c88)
3 Subnets, 3 Availability Zones

After a database is created, you can't change its VPC. Only VPCs with a corresponding DB subnet group are listed.

my-vpc (vpc-06eb8795e39f55c88)

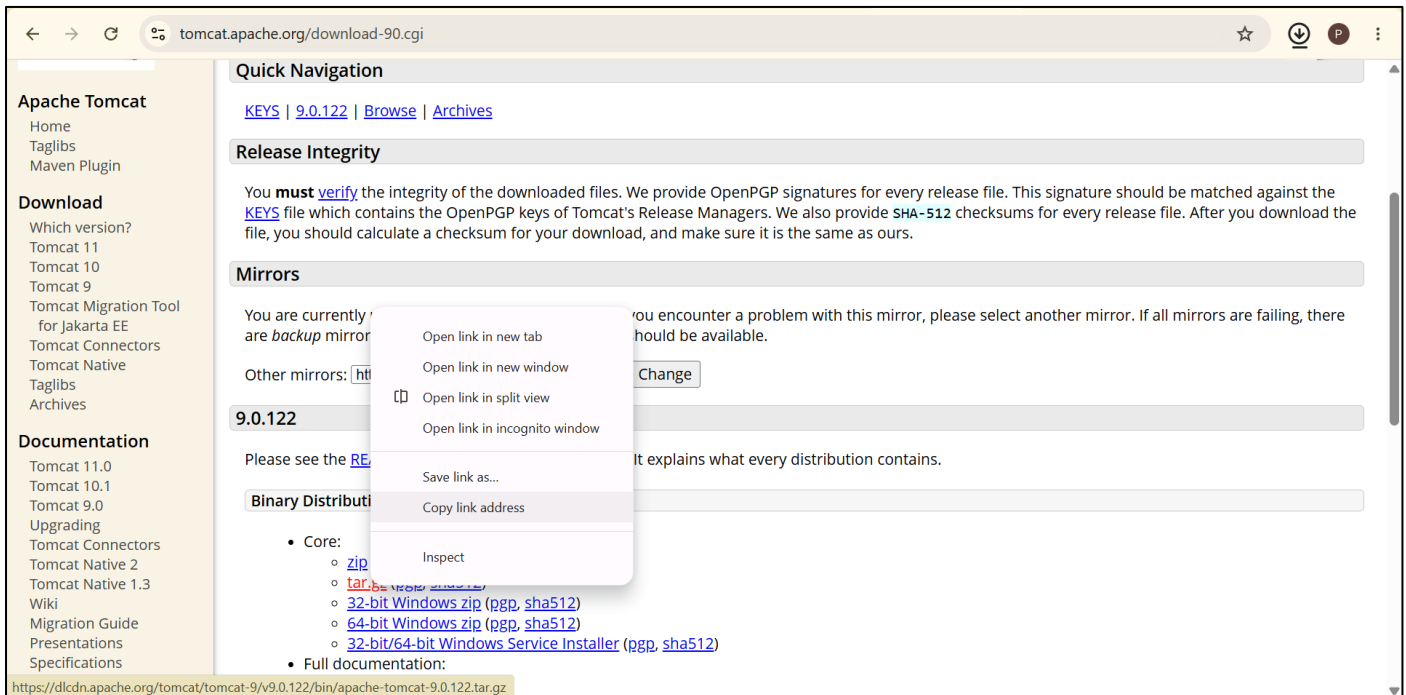
```
[ec2-user@jump-server ~]$ ssh -i 3-tier-key.pem ec2-user@10.0.27.244
```

```
#####  
Amazon Linux 2023  
  
#####  
https://aws.amazon.com/linux/amazon-linux-2023  
  
Last login: Sat Oct 3 09:31:03 2026 from 10.0.8.150  
[ec2-user@application-server ~]$ sudo -i  
[root@application-server ~]# yum install java -y  
Last metadata expiration check: 0:12:57 ago on Sat Oct 3 09:21:53 2026.  
Dependencies resolved.
```

Package	Architecture	Version	Repository	Size
Installing:				
java-26-amazon-corretto	x86_64	1:26.0.2+11-1.amzn2023.1	amazonlinux	207 k
Installing dependencies:				
alsa-lib	x86_64	1.2.7-2-1.amzn2023.0.4	amazonlinux	503 k
cairo	x86_64	1.18.0-4.amzn2023.0.3	amazonlinux	717 k
dejavu-sans-fonts	noarch	2.37-16.amzn2023.0.2	amazonlinux	1.3 M
dejavu-sans-mono-fonts	noarch	2.37-16.amzn2023.0.2	amazonlinux	467 k
dejavu-serif-fonts	noarch	2.37-16.amzn2023.0.2	amazonlinux	1.0 M

18. Download and Install Apache Tomcat

Searched for **Apache Tomcat 9** on the official Apache website, copied the **link address**, and downloaded it on the **application-server**. The Tomcat package was then extracted into the **/opt** directory to prepare the server for application deployment.



```
[root@application-server ~]# curl -O https://d1cdn.apache.org/tomcat/tomcat-9/v9.0.122/bin/apache-tomcat-9.0.122.tar.gz
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left   Speed
100 12.58M 100 12.58M    0     0  232.9M      0  0:00:03  0:00:03 --:--:-- 232.9M
[root@application-server ~]# ls
apache-tomcat-9.0.122.tar.gz
[root@application-server ~]# tar -xvzf apache-tomcat-9.0.122.tar.gz -C /opt
apache-tomcat-9.0.122/conf/
apache-tomcat-9.0.122/conf/catalina.policy
apache-tomcat-9.0.122/conf/catalina.properties
apache-tomcat-9.0.122/conf/context.xml
```

19. Deploy Application and Configure Apache Tomcat

Downloaded the **student.war** application file and deployed it in the **webapps** directory of Apache Tomcat. The **MySQL Connector JAR** was added to the **lib** directory to enable database connectivity. Finally, the **context.xml** file was opened for configuring the database connection.

```
[ec2-user@application-server ~]$ sudo -i
[root@application-server ~]# cd /opt/
[root@application-server opt]# ls
apache-tomcat-9.0.122  aws
[root@application-server opt]# cd apache-tomcat-9.0.122/
[root@application-server apache-tomcat-9.0.122]# ls
BUILDING.txt  CONTRIBUTING.md  LICENSE  NOTICE  README.md  RELEASE-NOTES  RUNNING.txt  bin  conf  lib  logs  temp  webapps  work
[root@application-server apache-tomcat-9.0.122]# cd webapps/
[root@application-server webapps]# ls
ROOT  docs  examples  host-manager  manager  student.war
[root@application-server webapps]# cd ..
[root@application-server apache-tomcat-9.0.122]# ls
BUILDING.txt  CONTRIBUTING.md  LICENSE  NOTICE  README.md  RELEASE-NOTES  RUNNING.txt  bin  conf  lib  logs  temp  webapps  work
[root@application-server apache-tomcat-9.0.122]# cd lib/
[root@application-server lib]# ls
annotations-api.jar  catalina-tribes.jar  jasper.jar  tomcat-coyote-ffm.jar  tomcat-i18n-es.jar  tomcat-i18n-ru.jar  tomcat-util.jar
catalina-ant.jar  catalina.jar  jaspic-api.jar  tomcat-coyote.jar  tomcat-i18n-fr.jar  tomcat-i18n-zh-CN.jar  tomcat-websocket.jar
catalina-ha.jar  ecj-4.20.jar  jsp-api.jar  tomcat-dbcp.jar  tomcat-i18n-ja.jar  tomcat-jdbc.jar  websocket-api.jar
catalina-ssi.jar  el-api.jar  servlet-api.jar  tomcat-el-cs.jar  tomcat-i18n-ko.jar  tomcat-jni.jar
catalina-storeconfig.jar  jasper-el.jar  tomcat-api.jar  tomcat-i18n-de.jar  tomcat-i18n-pt-BR.jar  tomcat-util-scan.jar
[root@application-server lib]# curl -O https://s3-us-west-2.amazonaws.com/studentapi-cit/mysql-connector.jar
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left   Speed
100 983.8k 100 983.8k    0     0  451.8k      0  0:00:02  0:00:02 --:--:-- 451.8k
[root@application-server lib]# ls
annotations-api.jar  catalina-tribes.jar  jasper.jar  tomcat-api.jar  tomcat-i18n-de.jar  tomcat-i18n-pt-BR.jar  tomcat-util-scan.jar
catalina-ant.jar  catalina.jar  jaspic-api.jar  tomcat-coyote-ffm.jar  tomcat-i18n-es.jar  tomcat-i18n-ru.jar  tomcat-util.jar
catalina-ha.jar  ecj-4.20.jar  jsp-api.jar  tomcat-coyote.jar  tomcat-i18n-fr.jar  tomcat-i18n-zh-CN.jar  tomcat-websocket.jar
catalina-ssi.jar  el-api.jar  mysql-connector.jar  tomcat-dbcp.jar  tomcat-i18n-ja.jar  tomcat-jdbc.jar  websocket-api.jar
catalina-storeconfig.jar  jasper-el.jar  servlet-api.jar  tomcat-el-cs.jar  tomcat-i18n-ko.jar  tomcat-jni.jar
[root@application-server lib]# cd ..
[root@application-server apache-tomcat-9.0.122]# ls
BUILDING.txt  CONTRIBUTING.md  LICENSE  NOTICE  README.md  RELEASE-NOTES  RUNNING.txt  bin  conf  lib  logs  temp  webapps  work
[root@application-server apache-tomcat-9.0.122]# cd conf/
[root@application-server conf]# ls
catalina.policy  catalina.properties  context.xml  jaspic-providers.xml  jaspic-providers.xsd  logging.properties  server.xml  tomcat-users.xml  tomcat-users.xsd  web.xml
[root@application-server conf]# vim context.xml
```


22. Connect to Amazon RDS and Create Database

Started the **MariaDB** service, downloaded the Amazon RDS SSL certificate, and securely connected to the Amazon RDS MariaDB instance. After establishing the connection, created the **studentapp** database and verified the database by querying the students table.

```
[root@database-server ~]# systemctl start mariadb.service
[root@database-server ~]# systemctl enable mariadb.service
Created symlink /etc/systemd/system/mysql.service → /usr/lib/systemd/system/mariadb.service.
Created symlink /etc/systemd/system/mysqld.service → /usr/lib/systemd/system/mariadb.service.
Created symlink /etc/systemd/system/multi-user.target.wants/mariadb.service → /usr/lib/systemd/system/mariadb.service.
[root@database-server ~]# curl -o global-bundle.pem https://truststore.pki.rds.amazonaws.com/global/global-bundle.pem
  % Total    % Received % Xferd  Average Speed   Time    Time     Current
                                 Dload  Upload   Total   Spent    Left     Speed
100 166.0k 100 166.0k    0     0  200.6k      0      0     0      0      0     0
[root@database-server ~]# ls
global-bundle.pem
[root@database-server ~]# mysql -h database-1.c908qegi0j5p.ap-south-1.rds.amazonaws.com -P 3306 -u admin -p --ssl-verify-server-cert --ssl-ca=./global-bundle.pem
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 262
Server version: 11.8.8-MariaDB-log managed by https://aws.amazon.com/rds/

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]>
```

```
MariaDB [(none)]> show databases;
+-----+
| Database |
+-----+
| information_schema |
| innodb |
| mysql |
| performance_schema |
| studentapp |
| sys |
+-----+
6 rows in set (0.003 sec)

MariaDB [(none)]> use studentapp;
Database changed
MariaDB [studentapp]> show tables;
Empty set (0.001 sec)

MariaDB [studentapp]> CREATE TABLE if not exists students(student_id INT NOT NULL AUTO_INCREMENT,
-> student_name VARCHAR(100) NOT NULL,
-> student_addr VARCHAR(100) NOT NULL,
-> student_age VARCHAR(3) NOT NULL,
-> student_qual VARCHAR(20) NOT NULL,
-> student_percent VARCHAR(10) NOT NULL,
-> student_year_passed VARCHAR(10) NOT NULL,
-> PRIMARY KEY (student_id)
-> );
Query OK, 0 rows affected (0.035 sec)
```

23. Configure Nginx Reverse Proxy

Installed **Nginx** on the **Jump Server** and configured it as a reverse proxy. Modified the **nginx.conf** file to forward incoming HTTP requests to the Apache Tomcat Application Server running on port 8080. This setup allows users to access the application through the Jump Server's public IP while keeping the Application Server securely deployed in a private subnet.

```
[ec2-user@jump-server ~]$ sudo -i
[root@jump-server ~]# yum install nginx -y
Amazon Linux 2023 Kernel Livepatch repository                               899 kB/s | 89 kB    00:00
Dependencies resolved.
=====
Package                                Architecture      Version            Repository          Size
-----
Installing:
nginx                                  x86_64            1:1.30.5-1.amzn2023.0.1  amazonlinux         34 k
```

```
Complete!
[root@jump-server ~]# systemctl start nginx.service
[root@jump-server ~]# systemctl enable nginx.service
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/system/nginx.service.
[root@jump-server ~]#
```

```

Complete!
[root@jump-server ~]# systemctl start nginx.service
[root@jump-server ~]# systemctl enable nginx.service
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/system/nginx.service.
[root@jump-server ~]# cd /etc/
[root@jump-server etc]# ls
DIR_COLORS          chrony.d            environment         hostname            libuser.conf       nfsmount.conf      rc1.d              shadow             systemd
DIR_COLORS.lightbgcolor  chrony.keys        ethertypes         hosts              libuser.conf       nginx              rc2.d             shadow-            terminfo
GREP_COLORS          cifs-utils         exports            idmapd.conf        locale.conf         nginx              rc3.d             shells            tmpfiles.d
NetworkManager       cloud              exports.d          image-id           localtime          nsdswitch.conf    rc4.d             skel              trusted-key.key
X11                  cpupower-service.conf  filesystems       init.d             logrotate.conf     opt               rc5.d             ssh               udev
acpi                  cron.daily         fstab              inittab           logrotate.d        os-release        rc6.d             ssl               update-motd.d
adjtime              cron.hourly        gcrpypt            inputrc           magic              pam.d             request-key.conf  sssd              vimrc
aliases              cron.monthly       gnupg             kernel            man_db.conf        passwd            request-key.d     statetab.d        virrc
alternatives          cron.weekly        groff             keyutils          mke2fs.conf        passwd            resolv.conf       subgid            wgetrc
amazon               crontab            group             krb5.conf         modules-load.d     pki               rsyncd.conf      subuid            xattr.conf
amazon-linux-release  crypto-policies    group             ld.so.cache       motd               pm                rwtab.d          sudo.conf         yum.repos.d
amazon-linux-release-cpe  csh.cshrc         grub.d            ld.so.conf         motd               printcap          sas12            sudoers.d         xdg
at.deny               csh.login          grub2-efi.cfg     ld.so.conf.d       mtab               profile           screenrc         sysconfig          sysctl.conf
audit                 dbus-1             grub2.cfg         ld.so.conf.d       mtab               protocols         security          selinux           system-release
bash_completion.d     dnf                gssproxy          hibernate-config.cfg  libn1              libreport         rc.d              sestatus.conf     system-release-cpe
bashrc                dnsmasq            gssproxy          hibernate-config.cfg  libn1              libreport         rc.d              sestatus.conf     system-release-cpe
bindresvport.blacklist  dnsmasq            gssproxy          hibernate-config.cfg  libn1              libreport         rc.d              sestatus.conf     system-release-cpe
binfmt.d              dnsmasq            gssproxy          hibernate-config.cfg  libn1              libreport         rc.d              sestatus.conf     system-release-cpe
chkconfig.d           dnsmasq            gssproxy          hibernate-config.cfg  libn1              libreport         rc.d              sestatus.conf     system-release-cpe
chrony.conf            dnsmasq            gssproxy          hibernate-config.cfg  libn1              libreport         rc.d              sestatus.conf     system-release-cpe
[root@jump-server etc]# cd nginx/
[root@jump-server nginx]# ls
conf.d  fastcgi.conf  fastcgi_params  koi-utf  mime.types  nginx.conf  scgi_params  uwsgi_params  win-utf
default.d  fastcgi.conf.default  fastcgi_params.default  koi-win  mime.types.default  nginx.conf.default  scgi_params.default  uwsgi_params.default
[root@jump-server nginx]# vim nginx.conf

```

```

location / {
    proxy_pass http://privateIPOftomcat:8080/student/;
}

```

```

server {
    listen 80;
    listen [::]:80;
    server_name _;
    root /usr/share/nginx/html;

    # Load configuration files for the default server block.
    include /etc/nginx/default.d/*.conf;

    error_page 404 /404.html;
    location = /404.html {
    }

    error_page 500 502 503 504 /50x.html;
    location = /50x.html {
    }

    location / {
        proxy_pass http://10.0.27.244:8080/student/;
    }
}

```

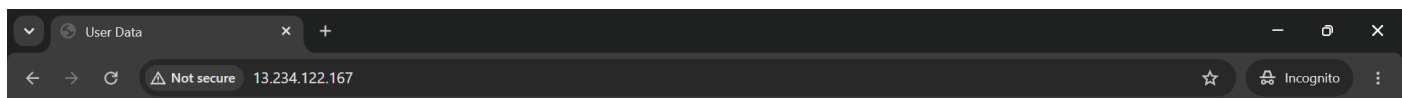
```

[root@jump-server nginx]# systemctl restart nginx.service
[root@jump-server nginx]#

```

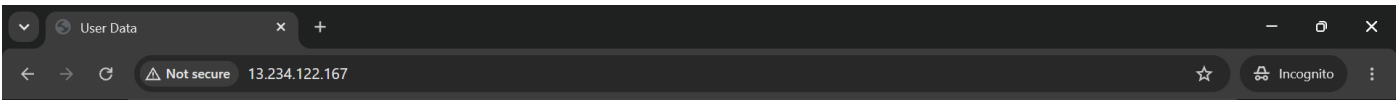
24. Verify Application Functionality and Database Connectivity

Verified the successful deployment of the three-tier application by accessing the Student Registration Form through the **Jump Server's public IP**. Entered sample student details and submitted the form successfully. After submission, verified that the data was displayed on the Students List page and confirmed that the same record was stored in the **MariaDB RDS** database using an SQL query. This validated the end-to-end communication between the Nginx Reverse Proxy, Apache Tomcat Application Server, and MariaDB RDS Database.



Student Registration Form

Student Name	
Student Address	
Student Age	
Student Qualification	
Student Percentage	
Year Passed	
register	



Student Registration Form

Student Name	Pooja
Student Address	Pune
Student Age	21
Student Qualification	Btech
Student Percentage	90
Year Passed	2026
register	

[Register Student](#)

Students List

Student ID	StudentName	Student Addr	Student Age	Student Qualification	Student Percentage	Student Year Passed	Edit	Delete
1	Pooja	Pune	21	Btech	90	2026	edit	delete

```
Database changed
MariaDB [studentapp]> show tables;
+-----+
| Tables_in_studentapp |
+-----+
| students              |
+-----+
1 row in set (0.001 sec)

MariaDB [studentapp]> select * from students;
+-----+-----+-----+-----+-----+-----+-----+
| student_id | student_name | student_addr | student_age | student_qual | student_percent | student_year_passed |
+-----+-----+-----+-----+-----+-----+-----+
| 1          | Pooja        | Pune         | 21          | Btech        | 90              | 2026                |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.002 sec)

MariaDB [studentapp]>
```

25. Configure Custom Domain and SSL Certificate

To make the application accessible through a custom domain instead of a public IP, a domain was purchased from Hostinger. The DNS was managed using Amazon Route 53, and an SSL/TLS certificate was issued using AWS Certificate Manager (ACM) to enable secure HTTPS communication.

aws [Logo] [Menu] [Search] [Alt+S] [Icons] [Asia Pacific (Mumbai)] [Pooja Daingade (736461510879)] [Pooja Daingade]

[AWS Certificate Manager](#) > [Certificates](#) > [Request certificate](#) > Request public certificate

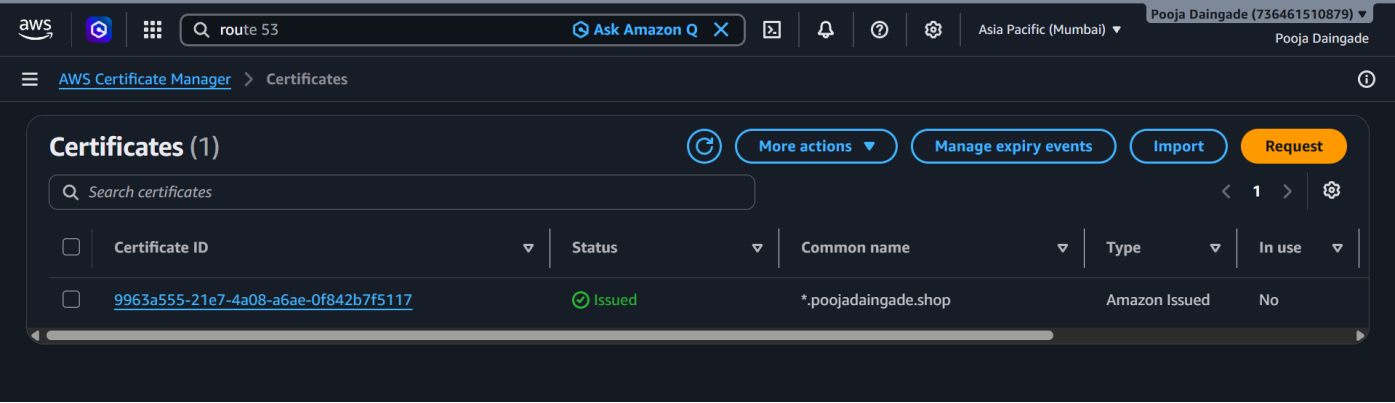
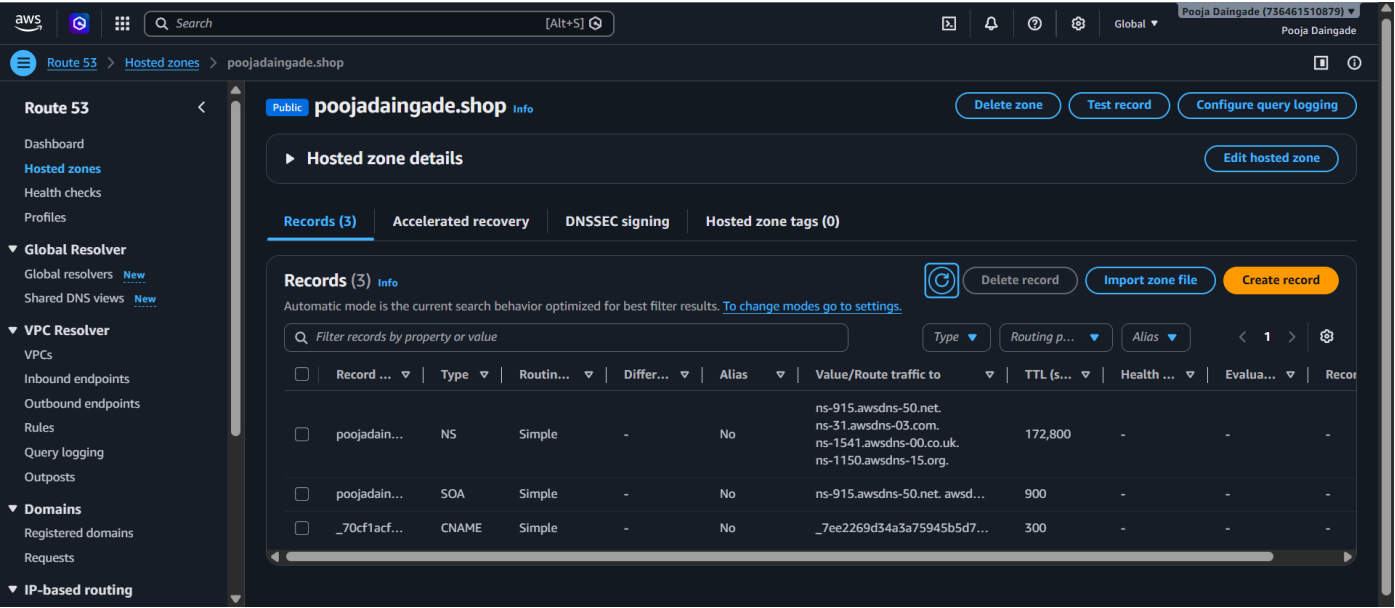
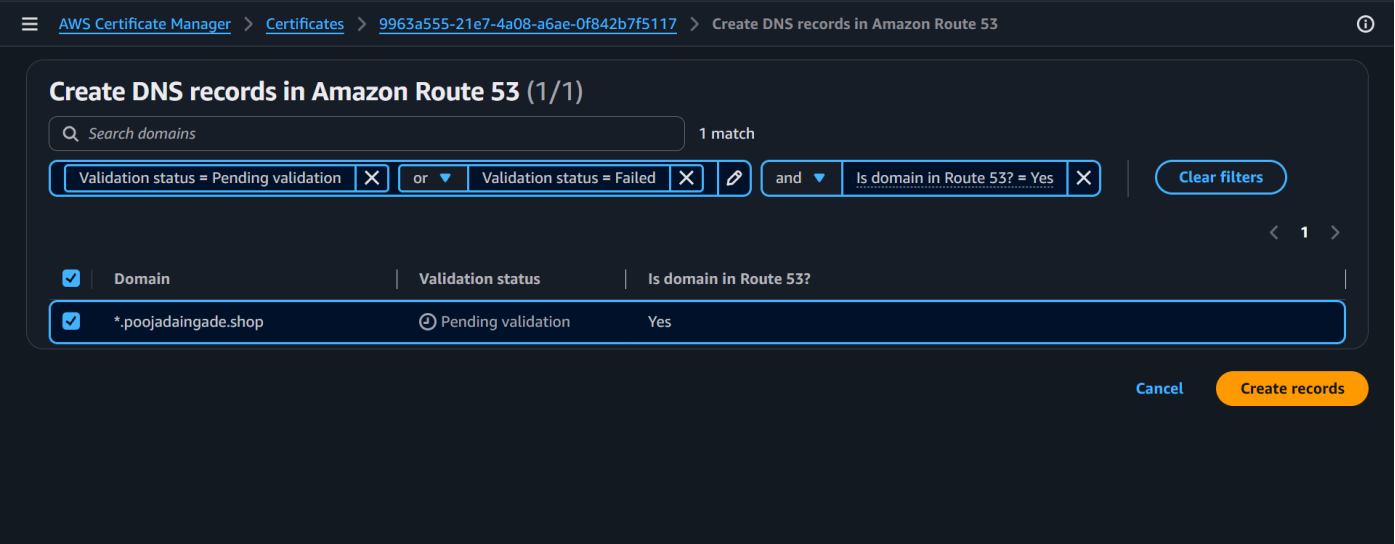
Request public certificate

Domain names
Provide one or more domain names for your certificate.

Fully qualified domain name [Info](#)

[Add another name to this certificate](#)

You can add additional names to this certificate. For example, if you're requesting a certificate for "www.example.com", you might want to add the name "example.com" so that customers can reach your site by either name.



aws [Search] [Alt+S] Ask Amazon Q Pooja Daingade (736461510879) Pooja Daingade

Route 53 > Hosted zones > poojadingade.shop > Create record

Create record Info

Quick create record Switch to wizard

Record 1 Delete

Record name Info

subdomain poojadingade.shop

Keep blank to create a record for the root domain.

Record type Info

A – Routes traffic to an IPv4 address and some AWS resources

Value Info

13.234.122.167

Enter multiple values on separate lines.

TTL (seconds) Info

300 1m 1h 1d

Recommended values: 60 to 172800 (two days)

Routing policy Info

Simple routing

Add another record

Cancel Create records

aws [Search] [Alt+S] Ask Amazon Q Pooja Daingade (736461510879) Pooja Daingade

Route 53 > Hosted zones > poojadingade.shop

Route 53 Public **poojadingade.shop** Info Delete zone Test record Configure query logging

Dashboard Hosted zones Health checks Profiles

Global Resolver New

Global resolvers Shared DNS views New

VPC Resolver

VPCs Inbound endpoints Outbound endpoints Rules Query logging Outposts

Domains

Registered domains Requests

Hosted zone details Info Edit hosted zone

Records (4) Accelerated recovery DNSSEC signing Hosted zone tags (0)

Records (4) Info Delete record Import zone file Create record

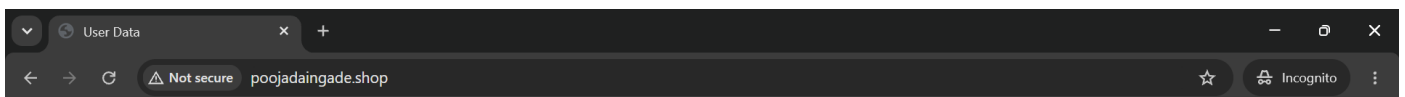
Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

Type Routing p... Alias

☐	Record name	Type	Routin...	Differ...	Alias	Value/Route traffic to	TTL (s...	Health ...	Evalu...	Recor...
☐	poojadingad...	A	Simple	-	No	13.234.122.167	300	-	-	-
☐	poojadingad...	NS	Simple	-	No	ns-915.awsdns-50.net. ns-511.awsdns-03.com. ns-1541.awsdns-00.co.uk. ns-1150.awsdns-15.org.	172,800	-	-	-
☐	poojadingad...	SOA	Simple	-	No	ns-915.awsdns-50.net. awsd...	900	-	-	-
☐	_70cf1acf9e8...	CNAME	Simple	-	No	_7ee2269d34a3a75945b5d7...	300	-	-	-

Result

The web application was successfully deployed and is accessible through the custom domain **poojadingade.shop**, replacing the EC2 public IP address. The application was tested successfully.



Student Registration Form

Student Name	
Student Address	
Student Age	
Student Qualification	
Student Percentage	
Year Passed	
register	

Project Outcomes

- Successfully deployed a secure AWS Three-Tier Web Application Architecture.
- Configured Amazon VPC, EC2, NAT Gateway, Route Tables, and Security Groups.
- Hosted the application using Apache Tomcat with Nginx as a reverse proxy.
- Integrated the application with Amazon RDS MariaDB for data storage.
- Configured the custom domain **poojadaingade.shop** using Amazon Route 53.
- Issued an SSL/TLS certificate using AWS Certificate Manager (ACM).
- Successfully verified end-to-end application functionality and database connectivity.

Conclusion

The AWS Three-Tier Web Application was successfully deployed on Amazon Web Services (AWS) using a secure and scalable architecture. The project demonstrated successful application deployment, database integration, custom domain configuration, and end-to-end functionality following AWS best practices.